

# Ejercicio Inverso Multiplicativo

Areli Alejandra Guevara Badillo

6CV4

miércoles, 17 de septiembre de 2025

04:04 p. m.

a) Calcula  $11^{-1} \bmod 24$

$$24 = 3 \cdot 2^3$$

$$11^{-1} = 11$$

$$\begin{aligned} 11 \cdot 11 &\bmod 24 \\ 121 &\bmod 24 = 1 \end{aligned}$$

$$\begin{array}{r} 2 \\ 11 \overline{) 24} \\ 2 \end{array}$$

$$\begin{aligned} 24 &= 11 \cdot 2 + 2 \\ 2 &= 24 + 11(-2) \end{aligned}$$

$$\begin{array}{r} 5 \\ 2 \overline{) 11} \\ 1 \end{array}$$

$$\begin{aligned} 11 &= 2 \cdot 5 + 1 \\ 1 &= 11 + 2(-5) \\ &= 11 + [24 + 11(-2)](-5) \\ &= 11 + 24(-5) + 11(10) \\ &= 11(11) + 24(-5) \end{aligned}$$

↑

b) Calcula  $12^{-1} \bmod 67$

$$67 = 67$$

$$\begin{aligned} 12^{-1} &= 28 \quad \rightarrow (12)(28) \bmod 67 \\ 336 &\bmod 67 = 1 \end{aligned}$$

$$\begin{array}{r} 5 \\ 12 \overline{) 67} \\ 7 \end{array}$$

$$\begin{aligned} 67 &= 12 \cdot 5 + 7 \\ 7 &= 67 + 12(-5) \end{aligned}$$

$$\begin{array}{r} 1 \\ 7 \overline{) 12} \\ 5 \end{array}$$

$$\begin{aligned} 12 &= 7 \cdot 1 + 5 \\ 5 &= 12 + 7(-1) \\ &= 12 + [67 + 12(-5)](-1) \\ &= 12 + 67(-1) + 12(5) \\ &= 12(6) + 67(-1) \end{aligned}$$

$$\begin{array}{r} 1 \\ 5 \overline{) 7} \\ 2 \end{array}$$

$$\begin{aligned} 7 &= 5 \cdot 1 + 2 \\ 2 &= 7 + 5(-1) \\ &= [67 + 12(-5)] + [12(6) + 67(-1)](-1) \\ &= 67 + 12(-5) + 12(-6) + 67(1) \\ &= 67(2) + 12(-11) \end{aligned}$$

$$\begin{array}{r} 2 \\ 2 \overline{) 5} \\ 1 \end{array}$$

$$\begin{aligned} 5 &= 2 \cdot 2 + 1 \\ 1 &= 5 + 2(-2) \\ &= 12(6) + 67(-1) + [67(2) + 12(-11)](-2) \\ &= 12(6) + 67(-1) + 67(-4) + 12(22) \\ &= 12(28) + 67(-5) \end{aligned}$$

$$\begin{array}{r} 2 \\ 1 \overline{) 2} \\ 0 \end{array}$$